

 Layer 3: API Gateways & Synchronization

- **Central ChatSync:** ChatService event bus routes messages from floating chatbots and AI drawers to target team chats live.
- **Express Backend API:** Service at `http://localhost:4001/api/ai` exposing `/chat`, `/agent`, `/confirm`, `/decline`.

 Layer 4: HRMs-AI Engine & HITL Safety

- **RAG Policy Chatbot:** Hybrid BM25 FullText + Cosine Similarity vector search over 3072-dim embeddings.
- **Action Agent:** ReAct function tool calling (`createTicket`, `assignTicket`, `applyLeave`, `sendMessageToUserOrChannel`, `queryTeamAttendance`).
- **HITL Risk Evaluation:** Staged in `agent_pending_actions` table with interactive UI confirmation cards.

3. Advanced Technology Roadmap

Technology	Capability	Implementation Goal
 Vision AI OCR	Multimodal Vision Analysis	Auto-extract expense receipt line items, medical bills, and ID proofs into reimbursement tickets.
 Multi-Agent Swarm	Subagent Orchestration	Decouple AI into specialized subagents (*Payroll Agent*, *Attendance Audit Agent*, *IT Agent*) via Redis Pub/Sub.
 Real-Time SSE Streaming	Token-by-Token Delivery	Stream AI answers letter-by-letter with typing indicators and WebSockets push alerts.
 GraphRAG Knowledge Graphs	Hierarchy Graph Traversal	Integrate Neo4j to query organizational reporting structures and escalation trees.
 Dedicated Vector DB	HNSW Indexing	Migrate vector storage to Qdrant or pgvector for sub-millisecond similarity search across millions of chunks.
 On-Device Voice AI	Quantized SLM Execution	Embed Whisper models via WebAssembly inside browser for zero-latency offline speech recognition.

Note: This architecture document is synced with the production codebase. Open this HTML document directly in Chrome or Safari to view the live interactive Mermaid flowchart diagram and generate print-ready PDF reports.